

EXPLAINED EXAMPLE 3 – ENERGY CONSERVATION WITH PHASE CHANGE

① Initially, 0.454 kg of water at 420 K with a quality of 0.6 is contained in a piston-cylinder device. The piston is then slowly moved until the water is entirely converted into liquid. Heat transfer maintains the temperature at 420 K. Calculate:

- ② a) The change in volume in meters cubed
b) The work in kJ
c) The heat in kJ

Given : $M = 0.454 \text{ kg}$

① $T_1 = 420 \text{ K}$

$$x_1 = 0.6$$

$$x_2 = 0$$

$$T_2 = 420 \text{ K}$$

Find : ΔV

② W_2

$$Q_2$$

Assume : S.C.S.

: quasi-steady

: isothermal

: constant mass

a) State 1 : $T_1 = 420 \text{ K}$
① $x_1 = 0.6$ } Table D.1

$$v_1 = x_1 v_g + (1 - x_1) v_f = (0.6)(0.4252) + (0.4)(0.001087)$$

$$v_1 = 0.256 \text{ m}^3/\text{kg}$$

$$\rightarrow V_1 = v_1 M = (0.256)(0.454) \rightarrow V_1 = 0.116 \text{ m}^3$$

② State 2 : $T_2 = 420 \text{ K}$

$$x_2 = 0$$

$$v_2 = v_f = 0.001087 \text{ m}^3/\text{kg}$$

$$\rightarrow V_2 = v_2 M = (0.001087)(0.454) \rightarrow V_2 = 0.00049 \text{ m}^3$$

$$\Rightarrow \Delta V = V_2 - V_1 \rightarrow \boxed{\Delta V = -0.1155 \text{ m}^3}$$

b) $\textcircled{1}$ ${}_1\bar{W}_2 = \int_{v_1}^{v_2} P dv \rightarrow \text{inside vapor dome, isothermal} = \text{isobaric}$
 $\rightarrow \text{Table D.1}$
 $= P \int_{v_1}^{v_2} dv = P_{\text{sat}}(v_2 - v_1) = (0.437309 \times 10^6)(-0.1155)$

$$\boxed{{}_1\bar{W}_2 = -50552.9 \text{ J}}$$

c) First law: $M\Delta u = {}_1Q_2 - {}_1\bar{W}_2$

$\textcircled{2}$ ${}_1Q_2 = M\Delta u + {}_1\bar{W}_2$

State 1: $u_1 = x_1 u_g + (1-x_1) u_f$
 $= (0.6)(2556.2) + (0.4)(618.13) = 1780.97 \frac{\text{kJ}}{\text{kg}}$

State 2: $u_2 = u_f = 618.13 \frac{\text{kJ}}{\text{kg}}$

$\Rightarrow M(u_2 - u_1) = (0.454)(618.13 - 1780.97) = -527.93 \text{ kJ}$

$\Rightarrow {}_1Q_2 = M\Delta u + {}_1\bar{W}_2$

$= -527.93 - 50.5529$

$$\boxed{{}_1Q_2 = -578.5 \text{ kJ}}$$